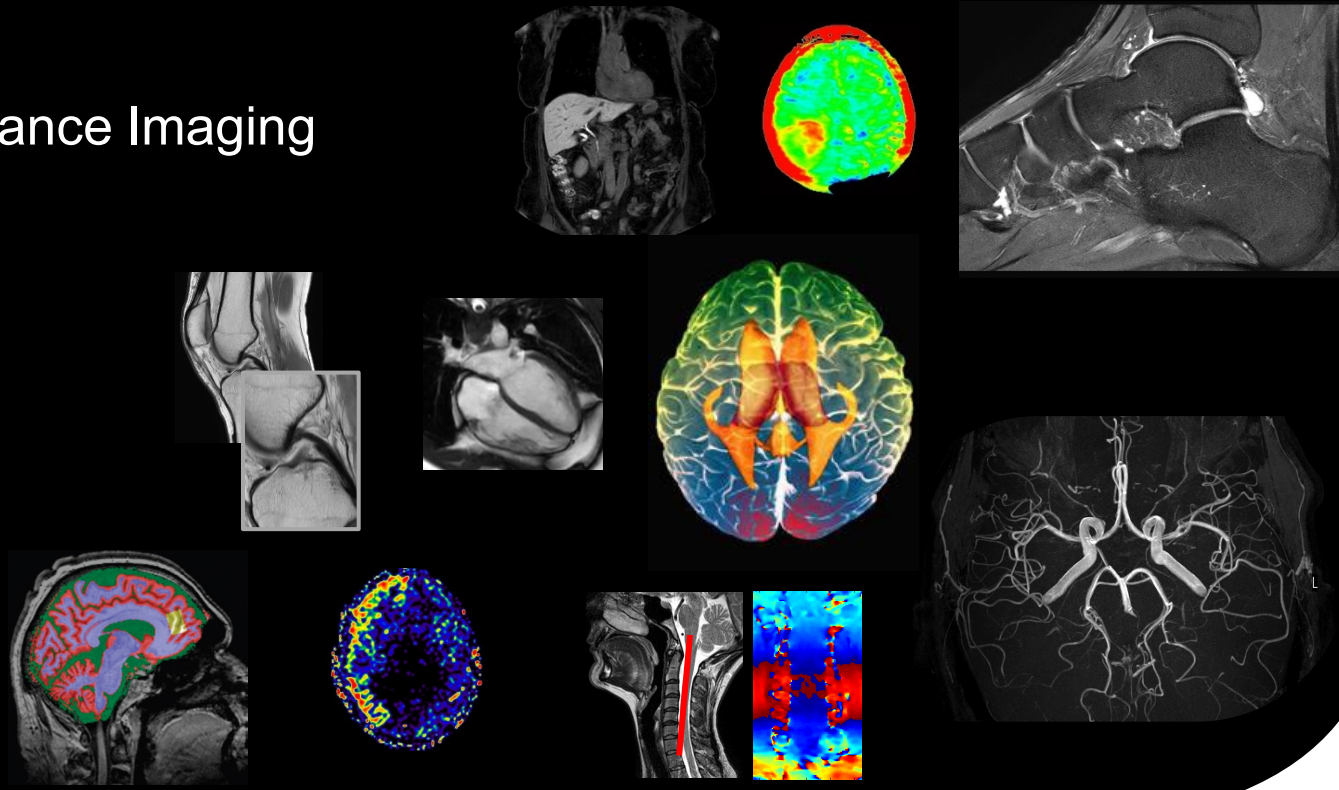
Introduction in Magnetic Resonance Imaging



Andreas Hock

1. Basics

- Introduction in Magnetic Resonance Imaging
- Principle

2. Contrast

- Relaxation
  - Longitudinal Relaxation
  - Transverse Relaxation
- De- and rephasing
- Contrast

3. MRI System Design

- Main magnetic field
- Gradient fields
- RF system

4. Spatial Encoding, Image Reconstruction

- Introduction
- Selective excitation
- Phase encoding
- Frequency encoding

5. MR Sequences

- Gradient echo
- EPI
- Spin-Echo
- TSE

6. Safety Aspects

- Main magnetic field
- Gradient field
- RF-field

7. Acceleration Techniques

- SENSE
- Compressed Sensing

8. Diffusion Weighted Imaging

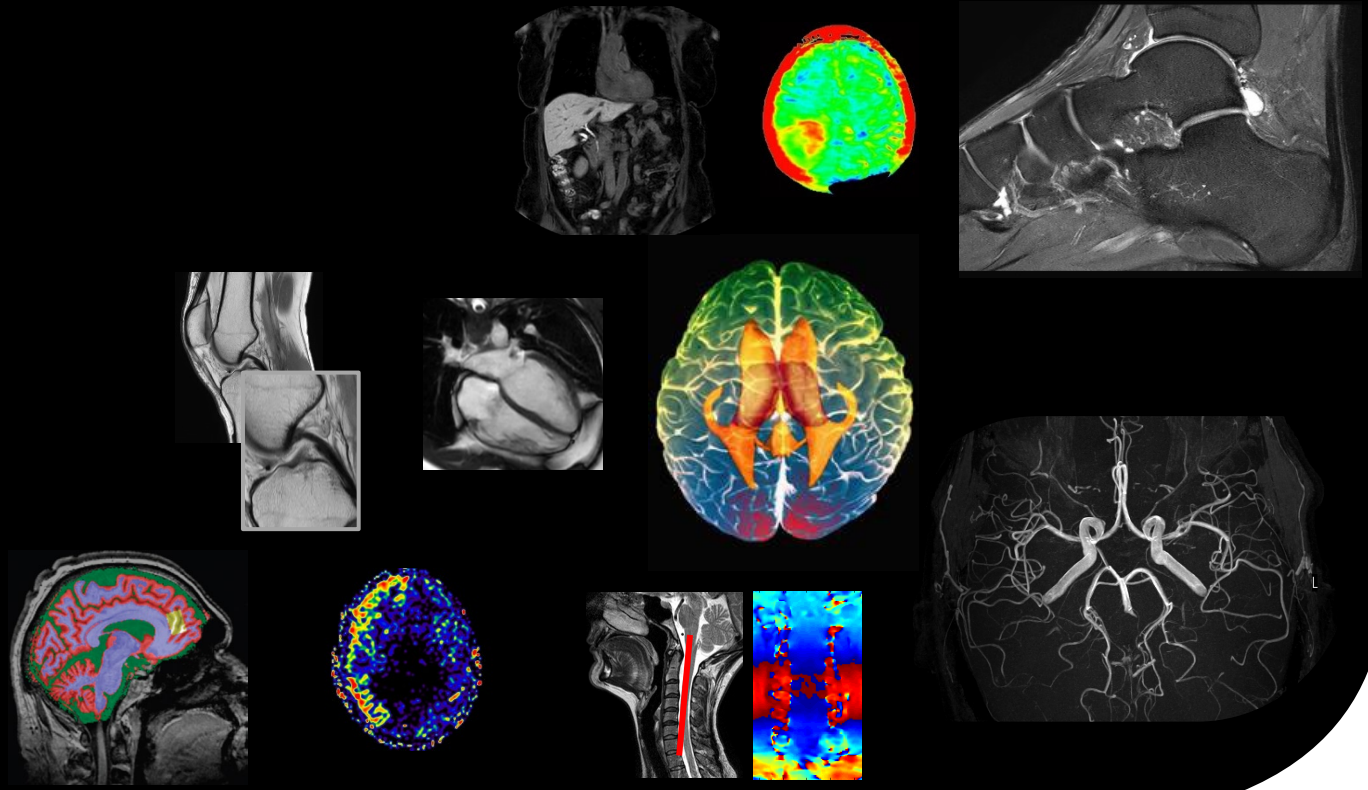
- Principle
- Clinical use
- Diffusion Tensor Imaging (DTI)

9. Functional MRI (fMRI)

- Principle
- Use cases
- Limitations

10. Prospects

- Future Development



Andreas Hock